

TWO RESIDUAL CASES OF SIMMONS'S POLYGONAL BIGRAPH CONJECTURE

ABSTRACT. Simmons conjectured that every polygonal bigraph $P_{m,k}$ is 2-edge-Hamiltonian. His addendum left the isomorphic pair $P_{34,9}$ and $P_{34,14}$ unresolved, while asserting the conjecture for all polygonal bigraphs on at most 66 vertices. Its residual table also omitted $P_{32,14}$, the only omitted entry not removed by the stated isomorphism reductions; as $P_{32,14}$ has 64 vertices, the 66-vertex assertion was not in fact established. We give five Hamilton cycles for each of $P_{34,9}$ and $P_{32,14}$ whose edge-pair coverage proves that both graphs are 2-edge-Hamiltonian. An explicit isomorphism settles $P_{34,14}$. Consequently the conjecture holds for every polygonal bigraph on at most 68 vertices, which both repairs the 66-vertex claim and extends it. For $P_{32,14}$ the conclusion is not new: it is the case $s = 4$ of the family $P_{8s,4s-2}$, treated in a companion note in this collection, and the certificate given here is an independent one. What is new is the class $P_{34,9} \cong P_{34,14}$ together with the finite audit. This is a finite verification, not a proof for arbitrary m .

1. STATEMENT

A graph is *2-edge-Hamiltonian* if every two distinct edges lie on a common Hamilton cycle. For $m \geq 3$ and $1 \leq k \leq \lfloor (m-1)/2 \rfloor$, let $P_{m,k}$ have bipartition

$$X = \{x_i : i \in \mathbb{Z}_m\}, \quad Y = \{y_i : i \in \mathbb{Z}_m\},$$

and edges, with subscripts modulo m ,

$$a_i = x_i y_i, \quad b_i = x_i y_{i-1}, \quad c_i = x_i y_{i+k}.$$

Thus $P_{m,k}$ is the usual $2m$ -cycle together with the chords of offset $2k+1$.

Simmons conjectured that every $P_{m,k}$ is 2-edge-Hamiltonian [2]. His addendum identified $P_{34,9} \cong P_{34,14}$ as an unresolved isomorphism class [3]; the isomorphism reductions used there originate in [1]. The same addendum asserted the result through 66 vertices. Its displayed residual table, however, omitted the entries $P_{31,11}$, $P_{32,8}$, and $P_{32,14}$. The first two reduce to settled parameters, but $P_{32,14}$ does not, so the assertion for 66 vertices was not in fact established: $P_{32,14}$ has 64 vertices. Theorem 1 therefore both repairs the 66-vertex claim and extends it to 68.

For $P_{32,14}$ the conclusion of Theorem 1 is not new. That graph is the case $s = 4$ of the family $P_{8s,4s-2}$, every member of which is proved 2-edge-Hamiltonian in the companion note [4] in this collection; the certificate

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we give below is an independent one, and we include it so that the 66-vertex range is repaired within the present note. What is new here is the isomorphism class $P_{34,9} \cong P_{34,14}$, which lies in no family treated in [4], together with the finite audit of the residual parameters with $m \leq 34$.

Theorem 1. *The polygonal bigraphs*

$$P_{32,14}, \quad P_{34,9}, \quad P_{34,14}$$

are 2-edge-Hamiltonian.

Corollary 2. *Every polygonal bigraph on at most 68 vertices is 2-edge-Hamiltonian.*

2. A COVERING CERTIFICATE

For $S \subseteq \mathbb{Z}_m$ and $t \in \mathbb{Z}_m$, write

$$S + t = \{s + t : s \in S\}.$$

If

$$\mathbb{Z}_m = A \dot{\cup} B \dot{\cup} C,$$

define

$$M(A, B, C) = \{a_i : i \in A\} \cup \{b_i : i \in B\} \cup \{c_i : i \in C\}.$$

This set meets every vertex of X once. Its Y -endpoints are indexed by A , $B - 1$, and $C + k$; hence it is a perfect matching precisely when those three sets partition $\mathbb{Z}_m$.

Lemma 3. *Let $M_0, \dots, M_4$ be perfect matchings of a cubic graph G . Suppose each*

$$H_r = G - M_r$$

is a Hamilton cycle and, for every two distinct edges e, f , some M_r contains neither e nor f . Then G is 2-edge-Hamiltonian.

Proof. For that index r , both edges belong to the Hamilton cycle H_r . $\square$

3. THE TWO CERTIFICATES

3.1. The graph $P_{34,9}$. For $r = 0, \dots, 4$, take the following subsets of $\mathbb{Z}_{34}$ and set

$$A_r = \mathbb{Z}_{34} \setminus (B_r \cup C_r) :$$

r	B_r	C_r
0	$\emptyset$	$\mathbb{Z}_{34}$
1	$\mathbb{Z}_{34}$	$\emptyset$
2	$\{25, 26\}$	$\{8, 17, 24, 33\}$
3	$\{16, 24\}$	$\{7, 15, 23, 32\}$
4	$\{15, 23\}$	$\{6, 14, 22, 31\}$

Put

$$M_r = M(A_r, B_r, C_r), \quad H_r = P_{34,9} - M_r.$$

Directly from (3.1), the sets A_r , $B_r - 1$, and $C_r + 9$ partition $\mathbb{Z}_{34}$, so every M_r is a perfect matching.

The cycle H_0 is the outer 68-cycle. The cycle H_1 consists of all a - and c -edges; its successive X -indices differ by -9 , and

$$\gcd(34, 9) = 1.$$

Thus H_0 and H_1 are Hamilton cycles. The three traces in Appendix A show that H_2, H_3, H_4 are also Hamilton cycles.

Now H_0 contains every a - and b -edge, while H_1 contains every a - and c -edge. Only pairs $\{b_i, c_j\}$ remain. The sets B_2, B_3, B_4 are pairwise disjoint, as are C_2, C_3, C_4 . Hence a fixed b_i is absent from at most one of H_2, H_3, H_4 , and the same holds for a fixed c_j . At least one of these three cycles therefore contains both edges. Lemma 3 proves that $P_{34,9}$ is 2-edge-Hamiltonian.

3.2. The graph $P_{32,14}$. For $r = 0, \dots, 4$, take the following subsets of $\mathbb{Z}_{32}$ and set

$$B_r = \mathbb{Z}_{32} \setminus (A_r \cup C_r) :$$

r	A_r	C_r
0	$\emptyset$	$\mathbb{Z}_{32}$
1	$\mathbb{Z}_{32}$	$\emptyset$
2	$\{0, 16\}$	$\{1, 17\}$
3	$\{15, 31\}$	$\{0, 16\}$
4	$\{14, 30\}$	$\{15, 31\}$

Put

$$M_r = M(A_r, B_r, C_r), \quad H_r = P_{32,14} - M_r.$$

For every row, A_r , $B_r - 1$, and $C_r + 14$ partition $\mathbb{Z}_{32}$, so every M_r is a perfect matching.

The cycle H_0 is the outer 64-cycle. The cycle H_1 consists of all b - and c -edges; up to orientation, its successive X -indices differ by 15, and

$$\gcd(32, 15) = 1.$$

The remaining traces in Appendix A show that H_2, H_3, H_4 are Hamilton cycles.

Here H_0 contains every a - and b -edge, while H_1 contains every b - and c -edge. Only pairs $\{a_i, c_j\}$ remain. The sets A_2, A_3, A_4 are pairwise disjoint, as are C_2, C_3, C_4 . The same three-cycle argument proves that every such pair lies on a Hamilton cycle. Thus $P_{32,14}$ is 2-edge-Hamiltonian.

4. ISOMORPHISM AND THE FINITE RANGE

Let primed symbols denote $P_{34,14}$. Define

$$\phi(x_i) = x'_{19i}, \quad \phi(y_i) = y'_{19i-1}.$$

Since 19 is a unit modulo 34, this is a bijection, and

$$\phi(a_i) = b'_{19i}, \quad \phi(b_i) = c'_{19i}, \quad \phi(c_i) = a'_{19i}.$$

Indeed,

$$19(i-1) - 1 \equiv 19i + 14, \quad 19(i+9) - 1 \equiv 19i \pmod{34}.$$

Thus

$$P_{34,9} \cong P_{34,14},$$

completing the proof of Theorem 1.

For completeness, we record the finite audit needed for Corollary 2. Put

$$S_k = \{0, -1, k\} \subseteq \mathbb{Z}_m.$$

If $u \in \mathbb{Z}_m^\times$ and $d \in \mathbb{Z}_m$ satisfy

$$uS_k + d = S_\ell,$$

then

$$x_i \mapsto x'_{ui}, \quad y_j \mapsto y'_{uj+d}$$

is an isomorphism

$$P_{m,k} \cong P_{m,\ell}.$$

After Simmons's general results, the residual arithmetic condition is

$$2 < \gcd(m+1, k+1) < k+1.$$

This reduction is recorded in [2, 3]. For $m \leq 34$, its solutions and their reductions are exactly the following. An arrow

$$k \rightarrow \ell \ (u, d)$$

records $uS_k + d = S_\ell$.

m	residual parameters and reductions
14	$5 \rightarrow 3 \ (11, 0)$
19	$7 \text{ (settled in [2])}$
20	$5, 8 \text{ (settled in [2])}$
23	$8 \rightarrow 3 \ (20, 0)$
24	$9 \text{ (settled in [2])}$
26	$5 \rightarrow 4 \ (21, -1), 11 \rightarrow 6 \ (19, -1)$
27	$7 \rightarrow 4 \ (23, 0), 11 \rightarrow 5 \ (22, 0)$
29	$8 \rightarrow 12 \ (13, 12), 11 \rightarrow 12 \ (12, 12)$
31	$11 \rightarrow 12 \ (13, 12)$
32	$5 \rightarrow 13 \ (19, 0), 8 \rightarrow 7 \ (7, 7), 11 \rightarrow 3 \ (29, 0), 14$
34	$13 \rightarrow 12 \ (21, -1), 9 \rightarrow 14 \ (19, -1)$

Every arrow except $9 \rightarrow 14$ lands in a parameter covered by the results of [2]. The unreduced entry $P_{32,14}$ and the class

$$P_{34,9} \cong P_{34,14}$$

are precisely those proved in Theorem 1. Hence every $P_{m,k}$ with $m \leq 34$ is 2-edge-Hamiltonian. Since $P_{m,k}$ has $2m$ vertices, Corollary 2 follows.

APPENDIX A. HAMILTON TRACES

Each row below is a cyclic list of successive X -indices in the indicated spanning 2-factor. Consecutive entries, including the last and first, have a common Y -neighbor through two retained edges. Since each row contains every X -index exactly once, it certifies a Hamilton cycle.

$P_{34,9}$.

$$T_{34,2} = (0, 10, 20, 30, 6, 16, 25, 1, 11, 21, 31, 7, 17, 18, 28, 4, 14, \\ 24, 15, 5, 29, 19, 9, 8, 32, 22, 12, 2, 26, 27, 3, 13, 23, 33),$$

$$T_{34,3} = (0, 10, 20, 30, 6, 15, 5, 29, 19, 9, 33, 32, 22, 12, 2, 26, 16, \\ 17, 27, 3, 13, 23, 14, 4, 28, 18, 8, 7, 31, 21, 11, 1, 25, 24),$$

$$T_{34,4} = (0, 10, 20, 30, 6, 7, 17, 27, 3, 13, 22, 12, 2, 26, 16, 15, 25, \\ 1, 11, 21, 31, 32, 8, 18, 28, 4, 14, 5, 29, 19, 9, 33, 23, 24).$$

$P_{32,14}$.

$$T_{32,2} = (0, 14, 28, 10, 24, 6, 20, 2, 17, 3, 21, 7, 25, 11, 29, 15, \\ 16, 30, 12, 26, 8, 22, 4, 18, 1, 19, 5, 23, 9, 27, 13, 31),$$

$$T_{32,3} = (0, 18, 4, 22, 8, 26, 12, 30, 31, 13, 27, 9, 23, 5, 19, 1, \\ 16, 2, 20, 6, 24, 10, 28, 14, 15, 29, 11, 25, 7, 21, 3, 17),$$

$$T_{32,4} = (0, 18, 4, 22, 8, 26, 12, 30, 29, 11, 25, 7, 21, 3, 17, 31, \\ 16, 2, 20, 6, 24, 10, 28, 14, 13, 27, 9, 23, 5, 19, 1, 15).$$

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